C6.2 Sequencing transactions

每个master需要能处理多个并发请求，也需要能接收多个snoop请求

可能会存在并发请求

Many masters might issue transactions at the same time. The protocol permits each master to make multiple outstanding requests, and to receive multiple outstanding snoop transactions.

对所有的masetr设备都能进行公平的仲裁

It is the responsibility of the interconnect to ensure that there is a defined order in which transactions to the same cache line can occur, and that the defined order is the same for all components. In the case of two masters issuing transactions to the same cache line at approximately the same time, then the interconnect determines which of the transactions is sequenced first and which is sequenced last. The arbitration method used by the interconnect is not defined by the protocol.

ACE协议并没有规定仲裁的形式

The interconnect indicates the order of transactions to the same cache line by sequencing transaction responses and snoop transactions to the masters. The ordering rules are:

- if a master issues a transaction to a cache line and it receives a snoop transaction to the same cache line before it receives a response to the transaction it has issued, then the snoop transaction is defined as ordered first.
- if a master issues a transaction to a cache line and it receives a response to the transaction before it receives a snoop transaction to the same cache line, then the transaction issued by the master is defined as ordered first.

Note

The relative ordering of transaction responses and snoop transactions only applies to transactions to the same cache line.

这里所说的顺序都是针对相同CacheLine来说的

The interconnect must ensure the following:

- if the interconnect provides a master with a response to a transaction, it must not send that master a snoop transaction to the same cache line before it has received the associated **RACK** or **WACK** response from that master

如果Interconnect发送了一个响应给master，那么Interconnect在接收到RACK和WACK之前不能发送Snoop请求给这个master

- if the interconnect sends a snoop transaction to a master, it must not provide that master with a response to a transaction to the same cache line before it has received the associated **CRRESP** response from that master.

如果Interconnect向一个master发送另一个Snoop请求，那么在接收到Snoop请求的响应之前都不能响应master之前的请求

The diagram in Figure C6-1 shows that from the point that a master starts to receive a transaction response it is guaranteed not to receive a snoop transaction to the same cache line until it has asserted the acknowledge signal, indicating that the transaction has completed.

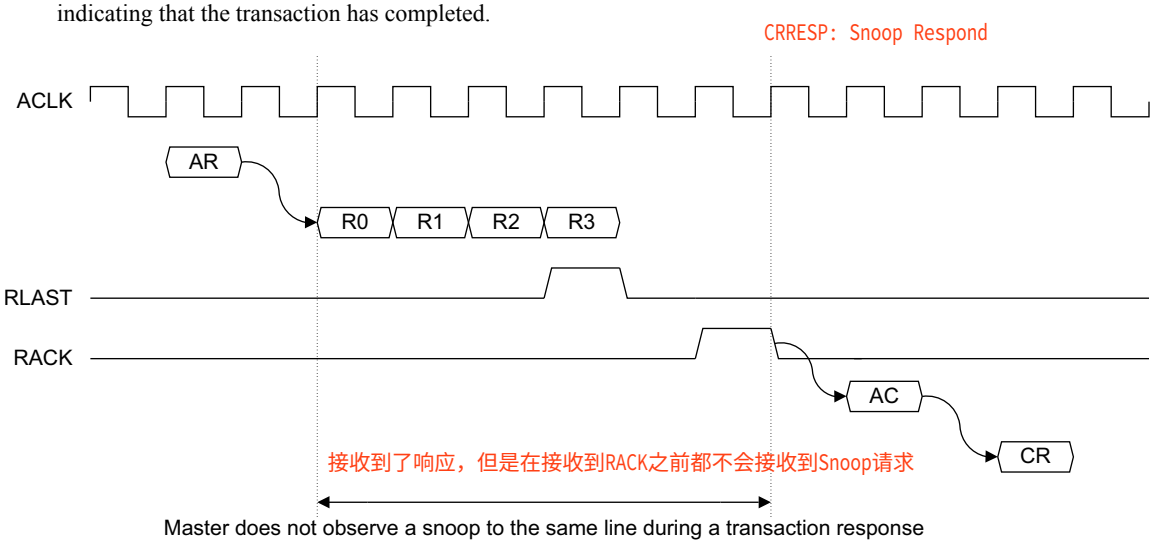


Figure C6-1 Transaction response before a snoop transaction

The diagram in Figure C6-2 on page C6-228 shows that if a master receives a snoop transaction to a cache line to which it has issued a transaction, but has not yet received a transaction response, then it is guaranteed not to see a transaction response until it has provided a snoop response.

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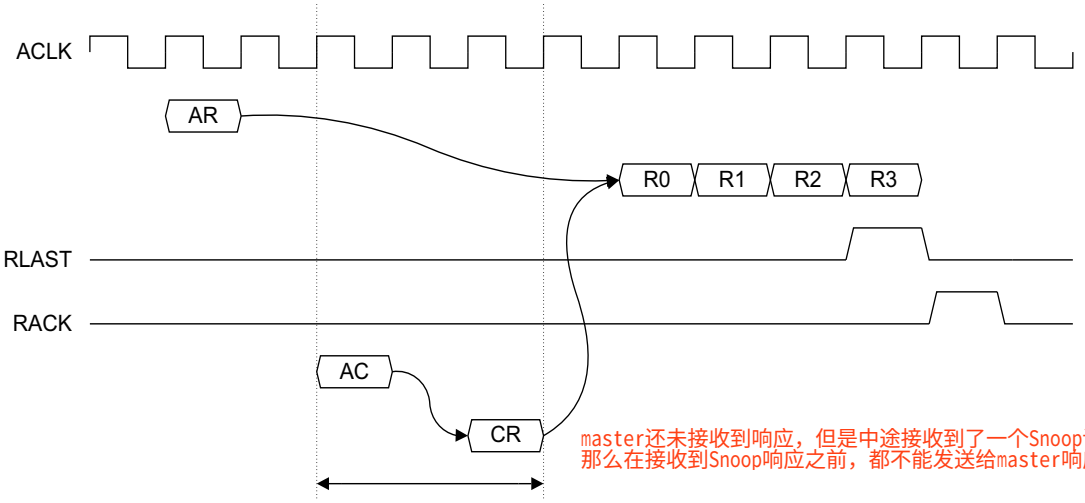


Figure C6-2 Snoop transaction before a transaction response

C6.2.1 Read and Write Acknowledge

RACK和WACK主要起到的是一个同步和标识传输结束的作用，因为ACE总线的互联系统中可能存在跨时钟域的情况

The read and write acknowledge signals are required to ensure correct operation where there is a delay between an interconnect and the master completing a transaction. For example, this can occur when a register or clock domain boundary exists between the interconnect and the master.

The master provides the read acknowledge signal **RACK** and the write acknowledge signal **WACK** to guarantee that the interconnect can determine when a transaction has completed at the master.

The master sends the **RACK** and **WACK** signals for all transactions, not only shareable transactions. This permits the signals to be generated using only the handshake signals on the read data channel or write response channel respectively.

The master must only send a read acknowledge after the last read data transfer in which **RLAST** is asserted. See *Read acknowledge signaling* on page C3-170. 在发送最后一个数据beat的时候，也就是RLAST拉高的时候，将RACK置高

The master must only send a write acknowledge after the write response handshake has occurred. See *Write Acknowledge signaling* on page C3-172. 在WriteRespond握手成功的时候，发送WACK

There is no mechanism to stall the **RACK** or **WACK** signal. The interconnect is required to accept the acknowledge in the same cycle as the master asserted the read or write acknowledge.

Interconnect需要在RACK和WACK置高的那一拍就准备好接收这一信息并且这两个请求是无法阻塞的

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